

Anna Natalie Chlopecki

PhD Candidate

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Education

Purdue University

Fall 2020 – Present

Doctor of Philosophy in Mathematics

Advisors: Uli Walther and Eric Samperton

University of Illinois at Urbana-Champaign

Fall 2016 – Spring 2020

Bachelor of Science in Mathematics *with High Distinction and Cum Laude*

Minor in Criminology, Law, and Society

College of Liberal Arts and Sciences James Scholar

Research Interests

Combinatorial Commutative Algebra

Determinantal ideals, Gröbner bases, Hilbert function and polynomial, matrix Schubert varieties

Computational Algebra

Computational complexity, counting complexity, group cohomology, knot theory

Awards

Purdue Carl Cowen Exceptional Promise Award	Spring 2026
Purdue Graduate School Summer Research Grant	Summer 2025
Purdue Department of Mathematics Barbara A. Kunze Award	Spring 2025
Purdue “Make a Difference” Bravo Award	Spring 2024
Purdue Graduate School Ross Fellowship	Fall 2020 – 2024
JMM 2020 MAA Undergraduate Student Poster Session Honorable Mention	Spring 2020
UIUC Lois Lackner Mathematics Scholarship	Spring 2018 and 2019
UIUC Elsie Thomas Fraser Mathematics Award	Spring 2017

Preprints

- (5) *Symmetric Bilinear Systems over $\mathbb{F}_2$ versus Symmetries of the Square*. With Eric Samperton. In preparation.
- (4) *Elimination Templates in Macaulay2*. With Manav Batavia, Cheng Chen, Timothy Duff, William Huang, Aolong Li, and Wanchun Shen. Submitted 2026. [arXiv:2605.00278](https://arxiv.org/abs/2605.00278).
- (3) *Antidiagonal Initial Complexes of Infinite Matrix Schubert Varieties are Cohen-Macaulay*. With Nathaniel Gallup and Jason Meintjes. Submitted 2026. [arXiv:2601.02612](https://arxiv.org/abs/2601.02612).
- (2) *An elementary proof that linking problems are hard*. With Shannon Cheng, Saarah Nazar, and Eric Samperton. Submitted 2025. [arXiv:2509.13120](https://arxiv.org/abs/2509.13120).
- (1) *Permutation Polynomials: Iteration of Shift and Inversion Maps over Finite Fields*. With Juliano Levier-Gomes, Wayne Peng, Alex Shearer, and Adam Towsley. 2019. [arXiv:1910.12928](https://arxiv.org/abs/1910.12928).

Software Development

EliminationTemplates

Release 1.26.05

Macaulay2 Package

Collaborators: Manav Batavia, Cheng Chen, Timothy Duff, William Huang, Aolong Li, and Wanchun Shen

External Talks and Presentations

Joint Mathematics Meetings 2027

Chicago, IL

Invited Speaker

January 12, 2027

Talk Title: Counting Complexity in Group Theory

External Talks and Presentations (Continued)

Fourth Workshop of the Kentucky-Ohio ALgebra Alliance (KOALA)

Presenter

Lexington, KY

May 11, 2026

Poster Title: Infinite Matrix Schubert Varieties are Cohen-Macaulay

AlGeCom XXVII

Presenter

West Lafayette, IN

April 11, 2026

Poster Title: Infinite Matrix Schubert Varieties are Cohen-Macaulay

Macaulay2 Workshop: Madison 2025 Lightning Talk

Speaker

Madison, WI

July 3, 2025

Talk Title: Investigating the Hilbertian Property

Purdue AWM Symposium

Invited Speaker

West Lafayette, IN

April 5, 2025

Talk Title: Investigating the Hilbertian Property

Joint Mathematics Meetings 2020

Presenter

Denver, CO

January 17, 2020

Poster Title: Monic Permutation Polynomials as Generators

North American School of Information Theory (NASIT) 2019

Presenter

Boston, MA

July 3, 2019

Poster Title: Entropy Vectors in Distributed Storage

Collaborators: Aubrey Laskowski and Weili Wu

Joint Mathematics Meetings 2019

Speaker

Baltimore, MD

January 19, 2019

Talk Title: How to: Create a Community for Undergraduate Mathematicians

Collaborator: Alice Chudnovsky

Champaign County History Museum

Presenter

Champaign, IL

March 8, 2018

Talk Title: The Four Color Theorem

Collaborators: Cora McVey, Tejo Notalapati, and Alan Zhou

Internal Talks and Presentations

Purdue Math Community of Practice Showcase

Presenter

West Lafayette, IN

April 30, 2026

Talk Title: AI Integration in Math Learning

Collaborators: Joseph Chen and Mahesh Sunkula

Purdue Ideal Conversations (Student Commutative Algebra Seminar)

Speaker

West Lafayette, IN

February 23, 2026

Talk Title: Infinite Matrix Schubert Varieties are Cohen-Macaulay

Purdue Student Colloquium Seminar

Speaker

West Lafayette, IN

December 10, 2025

Talk Title: Investigating the Hilbertian Property

Purdue Ideal Conversations (Student Commutative Algebra Seminar)

Speaker

West Lafayette, IN

April 28, 2025

Talk Title: The Local Hartshorne-Lichtenbaum Vanishing Theorem

Purdue Ideal Conversations (Student Commutative Algebra Seminar)

Speaker

West Lafayette, IN

November 25, 2024

Talk Title: Bumpless Pipe Dreams and Primary Decomposition

UIUC Undergraduate Friday Seminar

Speaker

Urbana, IL

November 8, 2019

Talk Title: Edelman Greene Tableaux and Complexity

Collaborator: Jackie Oh

Skills

Proficient in *Java*, *Python*, *SageMath*, and *Macaulay2*
Bilingual (English and Polish)

Collaborative Research and Coding Experience

M2@GT 2026: Building the Future of Macaulay2 Group Member <i>Project Focus: Public Relations and Communications</i> Faculty Leader: Gregory Smith (Queen's University)	Atlanta, GA May 17 – May 21, 2026
Macaulay2 Workshop: Madison 2025 Code Developer <i>Project Focus: Elimination Templates Package</i> Faculty Leader: Tim Duff (University of Missouri)	Madison, WI June 30 – July 4, 2025
Fields Institute Workshop on the Applications of Commutative Algebra Researcher <i>Project Title: Approximation Theory</i> Faculty Leaders: Mike DiPasquale (New Mexico State Univ.) and Nelly Villamizar (Swansea Univ.)	Toronto, ON May 26 – 30, 2025
Macaulay2, Computational Alg. Geometry and String Theory Workshop Code Developer <i>Project Focus: Normal Toric Varieties Package</i> Faculty Leader: Gregory Smith (Queen's University)	Salt Lake City, UT June 1 – 5, 2024

Research Mentorship

Undergraduate Initiative Graduate Student Mentor and Researcher <i>Project Focus: Determining the Computational Complexity of Counting Problems</i> Faculty Leader: Eric Samperton Undergraduate Researchers: Shannon Cheng (Purdue) and Saarah Nazar (UIUC)	West Lafayette, IN Summer 2025 – Present
Undergraduate Initiative Graduate Student Mentor and Researcher <i>Project Focus: Infinite Matrix Schubert Varieties</i> Faculty Leaders: Nathaniel Gallup and Jason Meintjes (Northeastern Univ.) Undergraduate Researchers: Claire MacDonald (Northeastern Univ.)	Virtual Summer 2025 and 2026
Purdue Experimental Mathematics Lab (PXML) Graduate Student Mentor <i>Project Title: Visualizing the Computational Complexity of Knots and Links</i> Faculty Leader: Eric Samperton Undergraduate Researchers: Shannon Cheng and Saarah Nazar	West Lafayette, IN Spring 2025

Teaching

Purdue Mathematics Department Instructor of Record <i>Linear Algebra (MA 26500)</i> <i>Applied Calculus I (MA 16010)</i> Recitation Instructor <i>Multivariate Calculus (MA 26100)</i> <i>Analytic Geometry and Calculus I (MA 16500)</i> Grader <i>Elements of Algebra I (MA 45300)</i> <i>Linear Algebra II (MA 35301)</i> <i>An Intro to Proof through Real Analysis (MA 30100)</i> <i>Diff. Eq. and Partial Diff. Eq. for Engineering and the Sciences (MA 30300)</i>	West Lafayette, IN Spring 2026 Fall 2024 and Spring 2026 Fall 2022 and Summer 2024 Fall 2020 Spring 2023, 2024, and 2025 Fall 2023 Fall 2023 Summer 2022
Freelance Tutor	West Lafayette, IN Summer 2023 – Summer 2024

Synergistic Activities and Engagement

Purdue Math Community of Practice Member	West Lafayette, IN Spring 2026
Purdue Bridge to Research Seminar Organizer	West Lafayette, IN Fall 2025 – Spring 2026
Purdue Association for Women in Mathematics Mathematics Graduate Student Mentor Vice President Secretary	West Lafayette, IN Fall 2021 – Present Fall 2025 – Spring 2026 Fall 2022 – Spring 2023
Purdue Bridge for Pursuing Higher Degrees (B4PhD) Scholars Program Organizer Mentor	Virtual Spring 2025 and 2026 Spring 2025
Purdue Sloan Center for Systemic Change Mathematics Department Ambassador	West Lafayette, IN Fall 2024 – Spring 2025
Purdue Mathematics Department Diversity Committee Graduate Liaison	West Lafayette, IN Fall 2024 – Spring 2025
Purdue Combinatorial Commutative Algebra Reading Group Organizer <i>Text Title: Combinatorial Commutative Algebra by Miller and Sturmfels</i>	West Lafayette, IN Spring 2024
Purdue Mathematics Department Graduate Representatives Graduate Representative	West Lafayette, IN Fall 2023 – Spring 2024
Purdue Student Colloquium Seminar Organizer	West Lafayette, IN Fall 2022 – Spring 2024
Purdue Women in Science Programs Mathematics and Statistics Departments' Liaison	West Lafayette, IN Fall 2021 – Spring 2023
Mastery Hour Learning Mathematics Volunteer Tutor	Virtual Spring 2021

Summer School and Workshop Participation

Mathematics Outreach Seminar and Training (MOST) Program at MoMath	New York, NY July 8 – 10, 2026
Fourth Workshop of the Kentucky-Ohio ALgebra Alliance (KOALA)	Lexington, KY May 11 – 12, 2026
11th Lake Michigan Workshop on Combinatorics and Graph Theory	Notre Dame, IN May 2 – 3, 2026
Summer CAMP (Commutative Algebra Market Preparation Workshop)	Lincoln, NE August 12 – 14, 2025
SLMath An Introduction to Recent Trends in Comm. Alg. Summer School	Toronto, ON June 2 – 13, 2025
SLMath Multi. and Diff. Graded Methods in Comm. Alg. Summer School	Moraga, CA July 29 – August 9, 2024
SLMath Recent Developments in Commutative Algebra Workshop	Berkeley, CA April 15 – 19, 2024
GEMS of Commutative Algebra Workshop	Minneapolis, MN November 10 – 12, 2023
SLMath Comm. Alg. and its Interaction with Alg. Geo. Summer School	Notre Dame, IN May 22 – June 2, 2023
Graduate Research Opportunities for Women (GROW)	Urbana, IL October 4 – 6, 2019

Graduate Conference Attendance

Joint Mathematics Meetings 2027	January 12 – 15, 2027
Geometry Labs United IV at the University of Wisconsin-Madison	June 1 – 4, 2026
AlGeCom XXVII at Purdue University	April 11, 2026
Algebra Across Time at Purdue University	April 26 – 27, 2025
Purdue AWM Symposium	April 5, 2025
Uwe Fest at the University of Notre Dame	August 12 – 16, 2024
Computational Algebraic Geometry and String Theory at the University of Utah	May 28 – 30, 2024
URiCA at the University of Nebraska-Lincoln	May 11 – 12, 2024
AMS Central Sectional Meeting at the University of Wisconsin-Milwaukee	April 20 – 21, 2024
Combinatorial Algebra meets Algebraic Combinatorics at LACIM	January 26 – 28, 2024
AlGeCom XXII at UIUC	October 22, 2022

Undergraduate Research Experience

Illinois Combinatorics Lab for Undergraduate Experiences (ICLUE)	Urbana, IL
Researcher	Fall 2017 – Spring 2020
<i>Project Title: An Investigation into Schubert Polynomial Parity</i>	
Faculty Mentor: Alexander Yong	

Brown University Summer@ICERM Mathematics REU	Providence, RI
Researcher	Summer 2019
<i>Project Title: Finite Field Statistics</i>	
Faculty Mentors: Adam Towsley (Rochester Inst. of Tech.) and Wayne Peng (Univ. of Rochester)	
Collaborators: Juliano Levier-Gomes and Alex Shearer	

Illinois Geometry Lab (IGL)	Urbana, IL
Researcher	Spring 2019
<i>Project Title: Bounds and Optimization for Distributed Storage</i>	
Faculty Mentors: Iwan Duursma, Xiao Li, and Hsin-Po Wang	
Collaborators: Aubrey Laskowski and Weili Wu	

Illinois Geometry Lab (IGL)	Urbana, IL
Researcher	Fall 2017
<i>Project Title: The Four Color Theorem: Archival Documentation and Outreach</i>	
Faculty Mentors: Jeremy Tyson, Chris Prom, and Bethany Anderson	
Collaborators: Xiaohan Liu, Cora McVey, Tejo Notalapati, and Alan Zhou	

Undergraduate Involvement

UIUC MATRIX Math Club	Urbana, IL
President	Fall 2017 – Spring 2020

UIUC Undergraduate Association for Women in Mathematics	Urbana, IL
MATRIX Liaison	Fall 2018 – Spring 2020
Mentorship Chair	Fall 2017 – Spring 2018

UIUC James Scholar Student Advisory Board	Urbana, IL
President	Fall 2019 – Spring 2020
Service Chair	Fall 2018 – Spring 2019
Service Committee Member	Fall 2017 – Spring 2018

Pi Mu Epsilon - Illinois Alpha Chapter	Urbana, IL
Member	Spring 2018 – Fall 2020

Past Work Experience

UIUC Campus Recreation - Ice Arena	Champaign, IL
Figure Skating Instructor	Fall 2017 – Spring 2020

Past Work Experience (Continued)

UIUC Computer Science Department	Urbana, IL
Course Assistant	
<i>Discrete Structures (CS 173)</i>	Fall 2018
Skokie Park District	Skokie, IL
Tennis Instructor	Summer 2014 – Summer 2016